

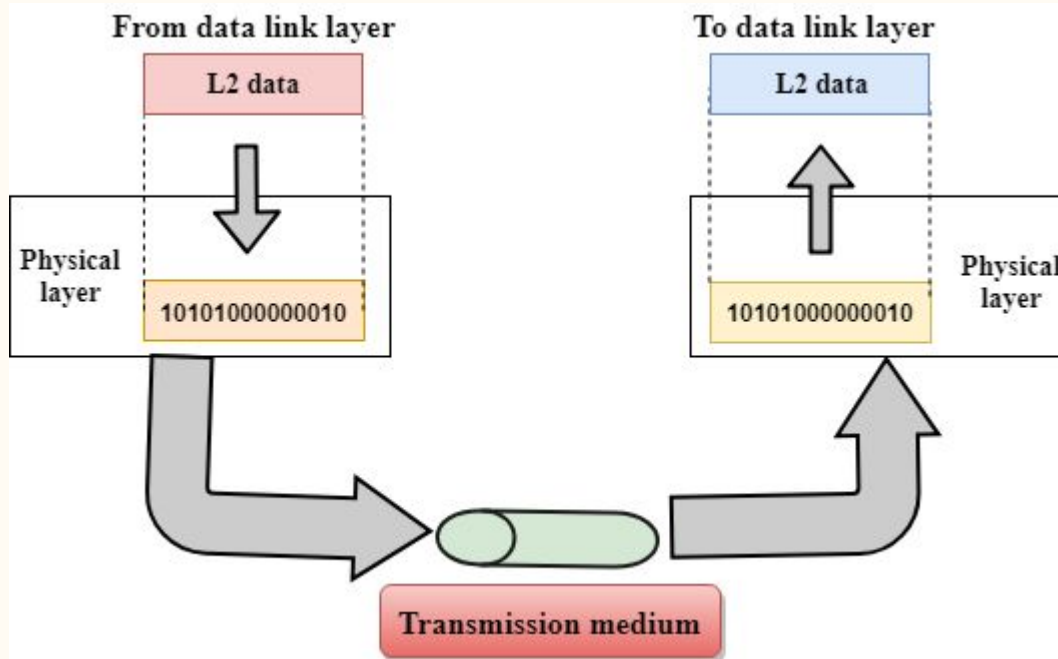
Physical Layer

Er. Sudarshan Devkota

What it does?

- The main functionality of the physical layer is to transmit the individual bits from one node to another node.
- It is the lowest layer of the OSI model.
- It establishes, maintains and deactivates the physical connection.

Function



Functions of Physical Layer

1. Representation of Bits:

The Physical Layer doesn't "understand" what a file or an image is. It only sees a raw stream of 0s and 1s. Its job is Signal Encoding or converting those logical bits into physical energy.

Working:

You must define a "Line Coding" scheme. For example, in Non-Return-to-Zero (NRZ), a high voltage (+5V) might be a "1" and a low voltage (0V) might be a "0".

Complex Encoding: Modern systems use more than just two levels to represent multiple bits with a single signal change.

Functions of Physical Layer

2. Transmission Rate/ Data Rate:

This function defines the number of bits sent per second, effectively determining the duration of each bit.

It refers to the bandwidth or capacity of a channel in bits per second (bps).

Example: A Fast Ethernet network has a maximum bandwidth of 100 Mbps, meaning it can transmit 100 million bits every second. The Physical Layer must be precise enough to create and detect a signal that short.

Functions of Physical Layer

3. Synchronization of Bits:

- The Physical Layer handles the bit-level synchronization between sender and receiver.
- This ensures the receiver samples the incoming signal at the exact correct intervals to correctly interpret each bit.
- This is achieved through clock recovery mechanisms.
- In asynchronous transmission, start and stop bits frame the data to manage timing.
- In synchronous transmission, a shared clock or self-clocking encoding schemes (like Manchester) are used to keep the devices' clocks aligned without a separate timing channel.

Functions of Physical Layer

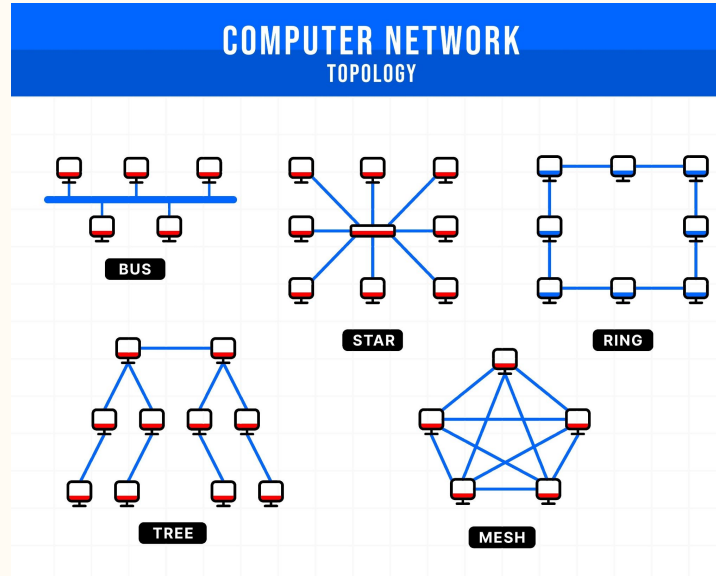
4. Interface and Line Configuration:

- This layer defines the transmission interface between devices and the medium, including the mechanical, electrical, and functional specifications of the connectors (pin assignments, impedance).
- It also specifies the line configuration, meaning how devices are connected to and share the medium
 1. *Point-to-Point*: A dedicated, exclusive link between two devices (e.g., a serial cable connecting a router to a console).
 2. *Multipoint (Multidrop)*: A single shared link connecting multiple devices, requiring medium access control handled by higher layers

Functions of Physical Layer

5. Physical Topology:

- It determines the physical arrangement or geometric layout of devices (nodes) on the network.



Functions of Physical Layer

6. Transmission Mode

- The Physical Layer defines the direction of signal flow between two connected devices.
- *Simplex*: Unidirectional communication. one device transmits, the other only receives (e.g., broadcast radio).
- *Half-Duplex*: Bidirectional communication, but only one direction at a time; the entire channel flips direction (e.g. a walkie-talkie).
- *Full-Duplex*: Simultaneous bidirectional communication. It requires separate transmit and receive paths or echo cancellation (e.g a telephone call, switched Ethernet).

Functions of Physical Layer

7. Multiplexing

- A method of sending multiple separate signals over a single physical line simultaneously to make efficient use of the available bandwidth.
- The Physical Layer combines multiple signals for simultaneous transmission over a single medium.
- In baseband, it uses Time-Division Multiplexing (TDM) where signals share the channel via time slots.
- In broadband, it uses Frequency-Division Multiplexing (FDM) where signals occupy different frequency bands simultaneously.

Data & Signal

Data: The information or message you want to send. It is the actual content—text, voice, video, or numbers. Example: The words you type in an email.

Signal: The physical form that data takes when it travels from one place to another. It is data converted into something that can move through wires, air, or glass. Example: The electrical pulses traveling through an Ethernet cable carrying your email.

Analog Signal

An analog signal is a continuous signal that varies smoothly with time and can take an infinite number of values within a given range.

$$s(t) = A \sin(2\pi ft + \phi)$$

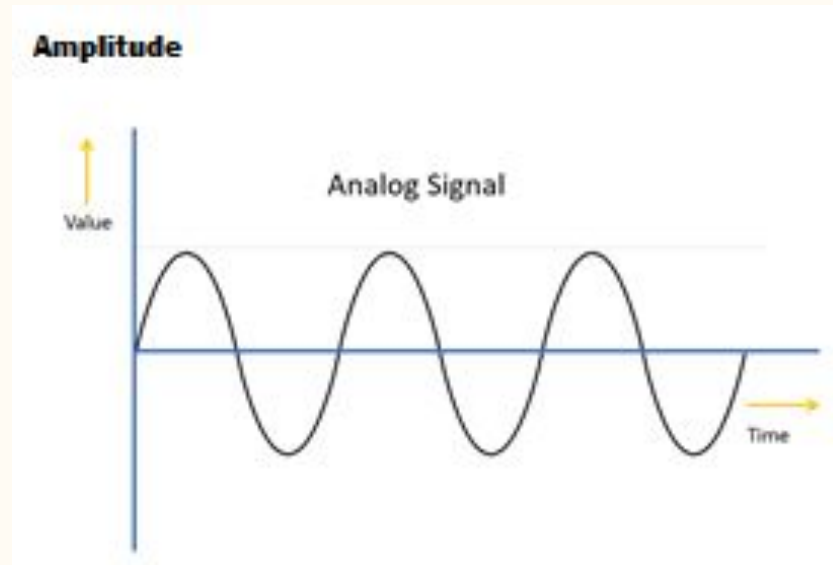
where,

A = Amplitude

f = Frequency

t = Time

ϕ = Phase



Analog Signal

1. Amplitude:

Amplitude refers to the maximum height or strength of the signal measured from the centerline of the waveform.

2. Frequency:

Frequency is the number of cycles completed by a signal in one second.

3. Phase:

Phase represents the position of the signal waveform relative to time zero. It indicates how much a signal is shifted horizontally.

Analog Signal

Advantages:

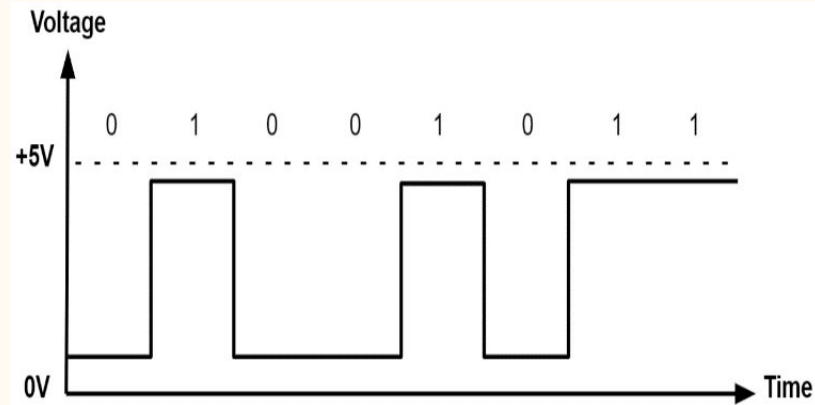
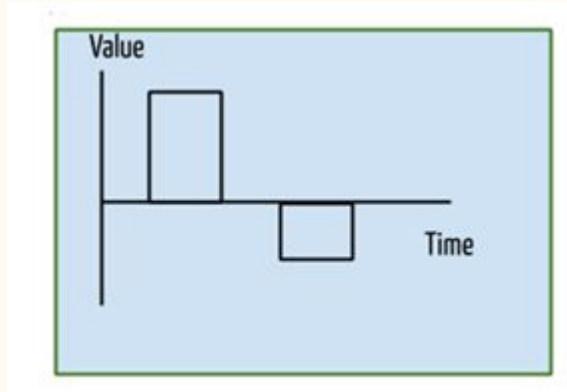
- Naturally represents real-world physical quantities
- Simple hardware implementation in some systems
- Smooth representation of information

Disadvantages:

- Highly affected by noise and interference
- Difficult to store and process accurately
- Quality decreases during transmission over long distances

Digital Signal

A digital signal is a signal that represents data using discrete levels rather than continuous values. It usually has only two possible states representing binary digits.



Digital Signal

Advantages:

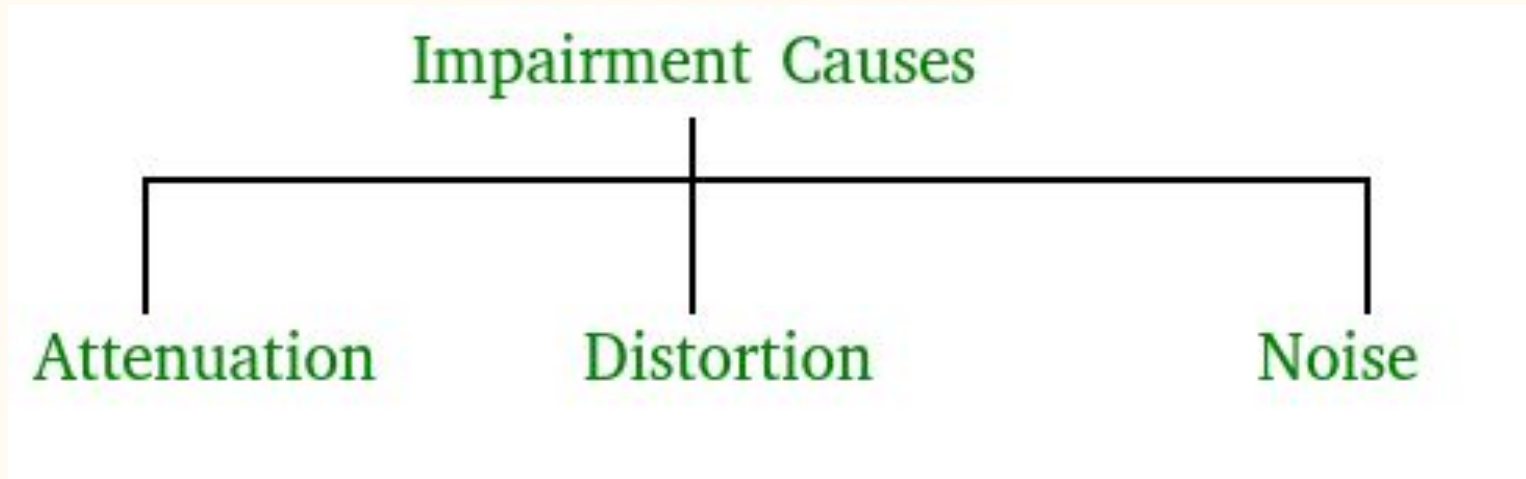
- Less affected by noise and interference
- Easy to store, process, and compress
- Supports error detection and correction
- Reliable for long-distance communication

Disadvantages:

- Requires more bandwidth in some cases
- Conversion from analog to digital may require additional devices (ADC/DAC)

Transmission Impairment

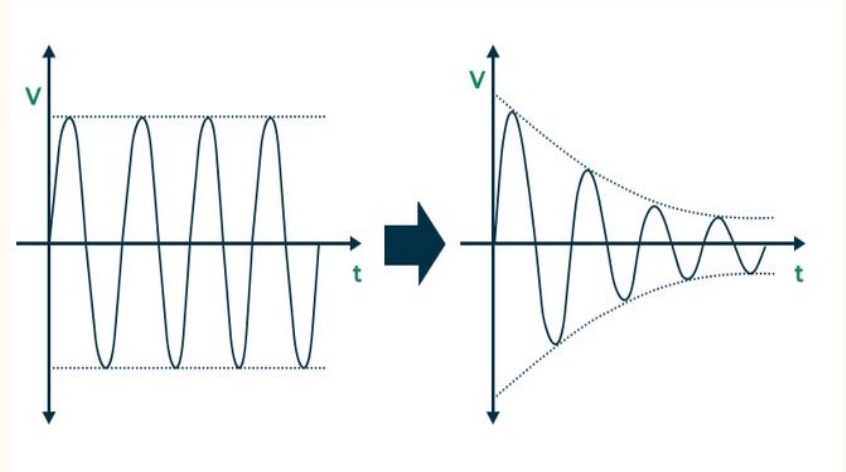
In a perfect world, the signal received would be identical to the signal sent. However, as signals travel through a medium, they encounter physical resistance and external interference. These are known as Impairments.



Attenuation

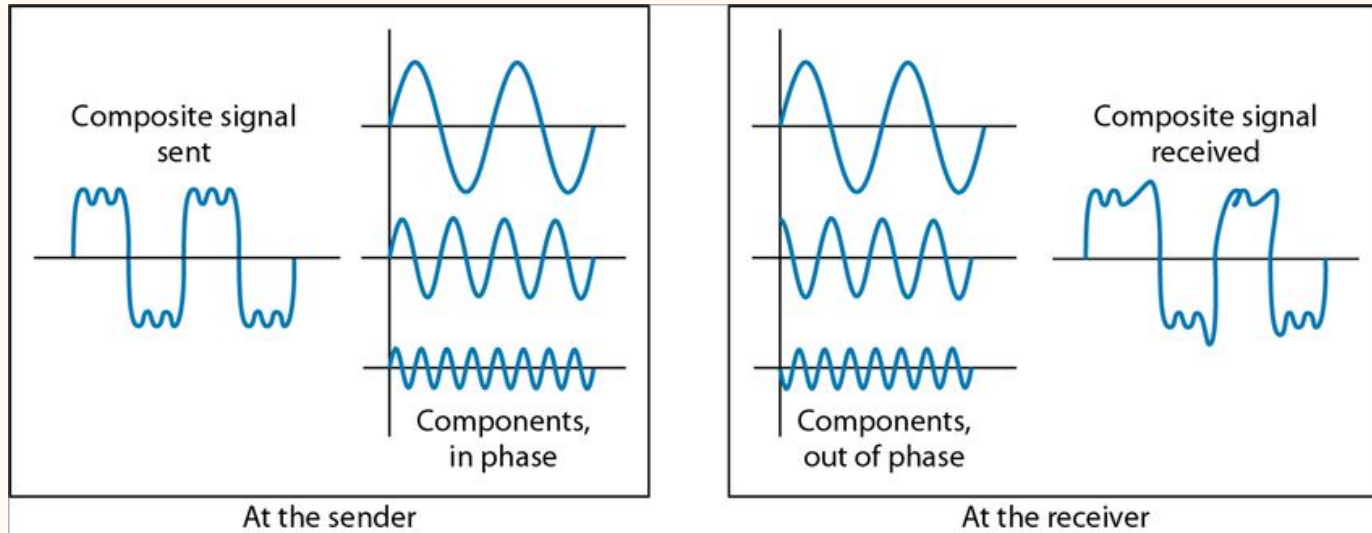
Attenuation is the loss of signal power as it travels through a medium. It occurs because the medium (like a copper wire) has resistance that converts electrical energy into heat. We use the Decibel (dB) to measure the relative strength of two signals or one signal at two different points

$$\text{Attenuation (dB)} = 10\log_{10} (P_2/P_1)$$



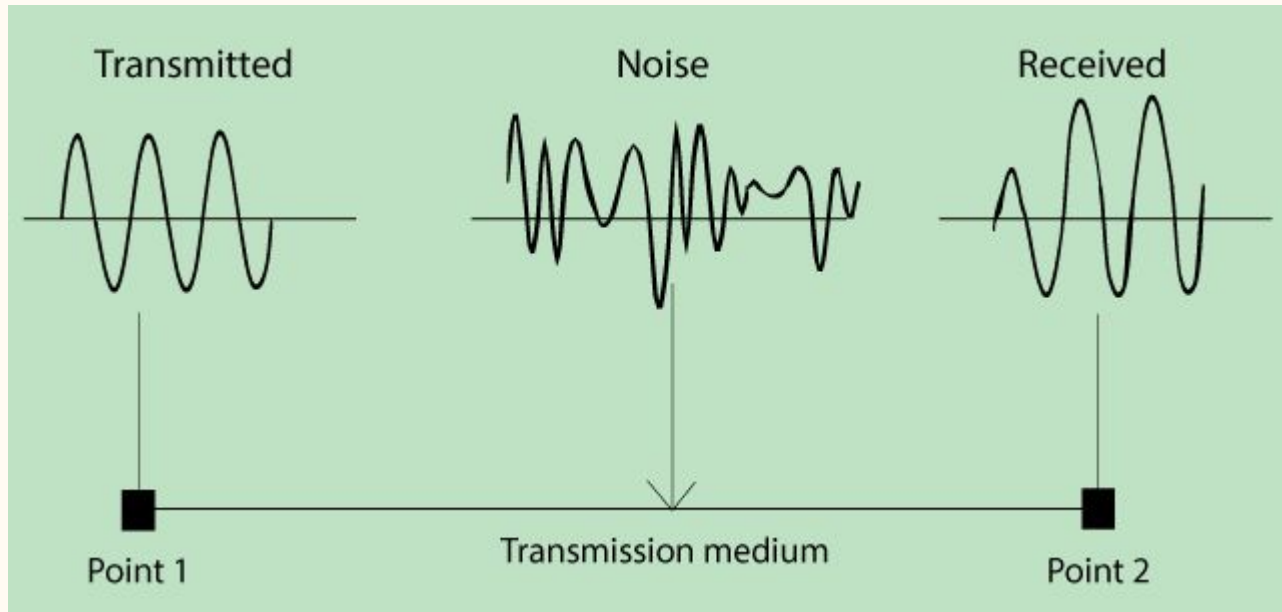
Distortion

Distortion occurs when the shape of a signal changes during transmission. The signal still reaches the receiver, but it is not exactly the same waveform that was transmitted. It occurs in composite signals.



Noise

Noise is any unwanted signal that gets added to the original signal during transmission.



Noise

1. Thermal Noise:

Thermal noise is the noise generated due to the random motion of electrons in a conductor caused by heat.

2. Induced Noise (Electromagnetic Interference)

Induced noise is caused when electromagnetic fields from external sources interfere with the signal in a transmission medium.

3. Cross Talk:

Crosstalk occurs when a signal transmitted on one communication channel interferes with another nearby channel.

Noise

4. Impulse

Impulse noise is irregular pulses or spikes(a signal with high energy in a very short period) generated by phenomena like that comes from power lines, lightning, spark due to loose contact in electric circuits and so on.

Data Rate

The data rate is a term to denote the transmission speed, or the number of bits per second transferred.

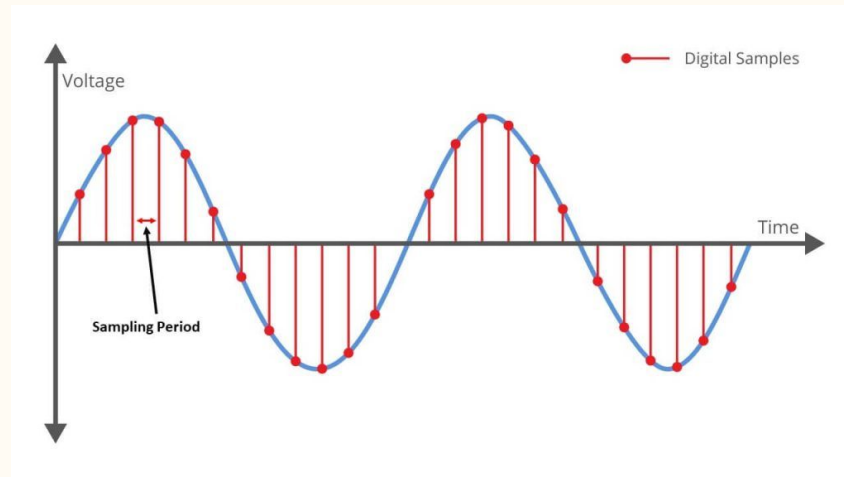
Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Sampling

An analog wave is continuous, meaning it has a value at every single quadrillionth of a second. A computer cannot store "infinite" time.

Sampling is the process of measuring the amplitude of a continuous analog signal at discrete, equal intervals of time.



Nyquist Sampling Theorem/Criteria

"To perfectly reconstruct a signal, the sampling rate must be at least twice the highest frequency component present in the original signal."

Formula: $F_s > 2 \times F_{max}$

F_s = Sampling Frequency

F_{max} = The highest frequency in the analog signal

The Nyquist Bit Rate (For Noiseless Channels)

The Formula: **BitRate = 2x Bandwidthx log₂(L)**

Bandwidth: The range of frequencies the cable can carry.

L: The number of signal levels used to represent data.

1. Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

2. Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12,000 \text{ bps}$$

Shannon Capacity (For Noisy Channels)

Formula: **Capacity = Bandwidth x $\log_2(1 + \text{SNR})$**

SNR (Signal-to-Noise Ratio): The ratio of signal power to noise power. A higher SNR means a "cleaner" line and a higher capacity.

Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint.

For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Shannon Capacity (For Noisy Channels)

- We can calculate the theoretical highest bit rate of a regular telephone line.
- A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications.
- The signal-to-noise ratio is usually 3162.
- For this channel the capacity is calculated as
$$\mathbf{C = B \log_2 (1 + SNR)}$$
$$= 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34,860 \text{ bps}$$
- This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Performance Metrics

1. Bandwidth:

Bandwidth in Hertz is the range of frequencies contained in a composite signal or the range of frequencies that a transmission medium can pass without significant loss of power.

A traditional telephone line is designed to carry frequencies between 300 Hz and 3,400 Hz. Therefore, the bandwidth of a telephone line is:

$$3400 \text{ Hz} - 300 \text{ Hz} = 3,100 \text{ Hz (or } 3.1\text{kHz)}.$$

Performance Metrics

1. Bandwidth:

Bandwidth in bits per second (bps) refers to the maximum theoretical data rate that a channel can support. It defines how many bits can be pushed through a point in the network in one second.

A "Fast Ethernet" cable has a digital bandwidth of 100 Mbps. This means, in a perfect world with no noise, the hardware can transmit 100 million bits every second.

Performance Metrics

2. Throughput:

Throughput is the measure of how fast data is actually being transferred through a point in the network at a specific moment. It is defined as the number of bits that pass through a network segment per unit of time (usually seconds).

Factors Affecting Throughput:

- Transmission Impairments
- Congestion
- Protocol Overhead
- Hardware Processing Power

Performance Metrics

3. Latency (Delay):

Latency is the total time required for an entire message to travel from the source to the destination.

It is the sum of four distinct delay components:

**Latency = propagation time + transmission time +
queuing time + processing delay**

Performance Metrics

3. Latency (Delay):

a. Propagation Time

Propagation time measures the time required for a bit to travel from the source to the destination.

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Performance Metrics

3. Latency (Delay):

B. Transmission Time

The time required to push all the bits of a message onto the wire

$$\text{Transmission Time} = \frac{\text{Message Size}}{\text{Bandwidth}}$$

Performance Metrics

3. Latency (Delay):

C. Queuing Time

The time a packet spends waiting in a router's buffer (memory) during high traffic.

D. Processing Time

The time a router takes to check for errors and decide the next path.

Performance Metrics

4. Jitter:

Jitter is the variation or fluctuation in the latency (delay) of packets belonging to the same data stream. It is the measure of the "unevenness" in packet arrival times.

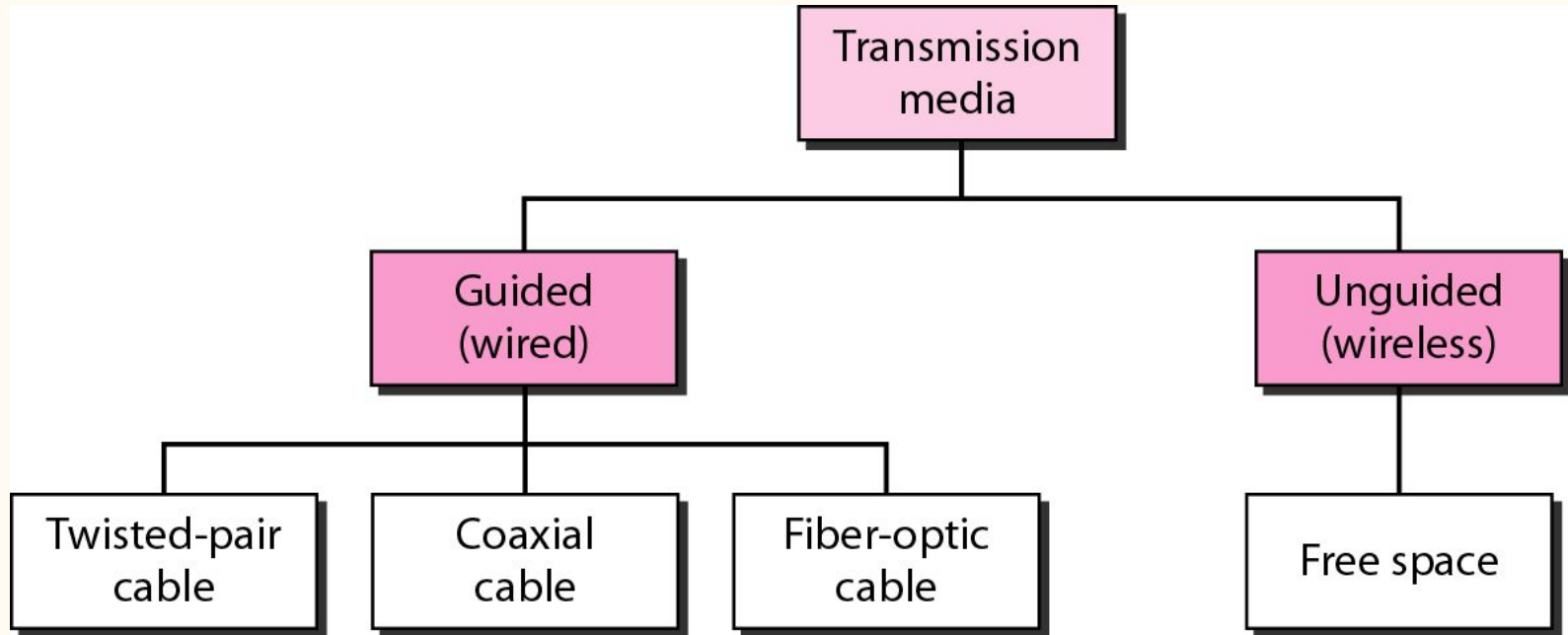
It occurs because different packets may take different paths through the network, or they may encounter varying amounts of Queuing Delay at different routers.

Performance Metrics

5. Bandwidth Delay Product:

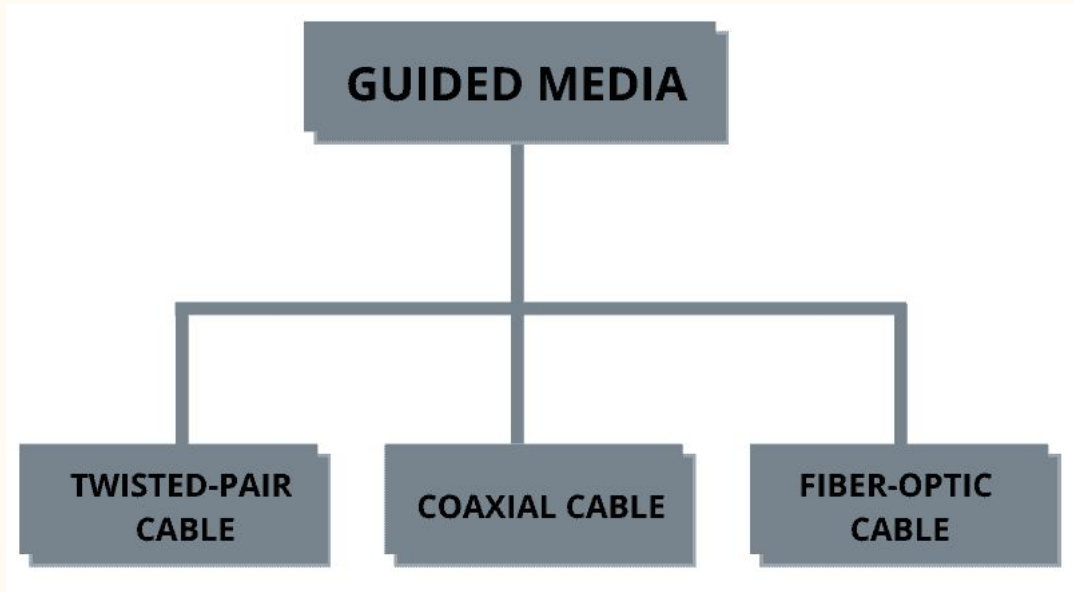
The Bandwidth-Delay Product is the mathematical product of the link's bandwidth (in bps) and its round-trip delay (in seconds). It represents the maximum number of bits that can be "in flight" on the transmission medium at any single moment.

Transmission Media



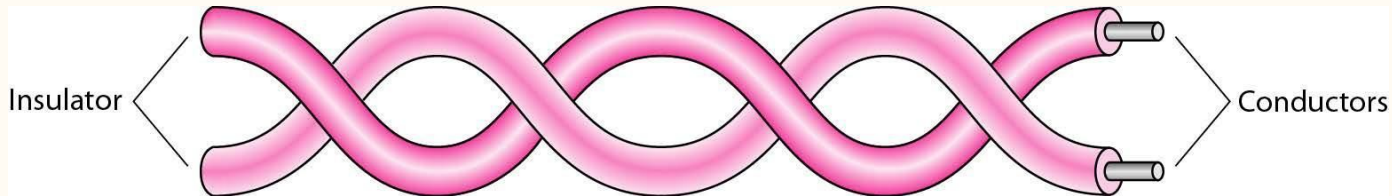
Guided Media

Guided transmission media, often referred to as bounded media, comprise the physical pathways through which data signals are directed and contained.



Guided Media: Twisted Pair

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- A signal is usually carried as the difference in voltage between the two wires in the pair. This
- provides better immunity to external noise because the noise tends to affect both wires the same, leaving the differential unchanged.



Guided Media: Twisted Pair

Advantages:

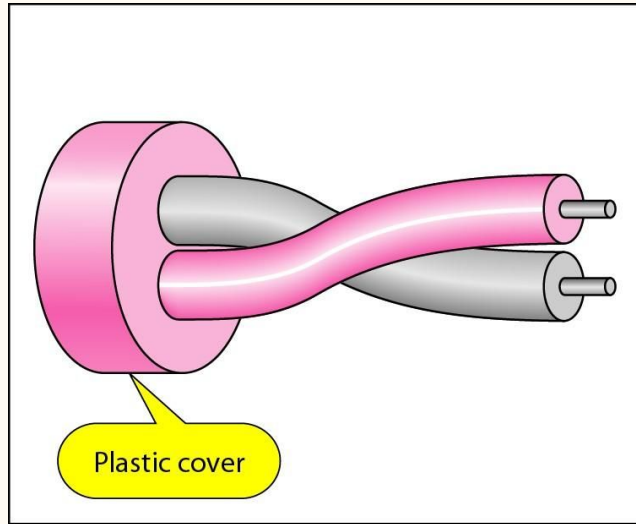
- Protect against crosstalk & interference.
- Easy to work with.
- Easy to add computers to network.
- Well-understood technology.
- Less expensive.

Disadvantages:

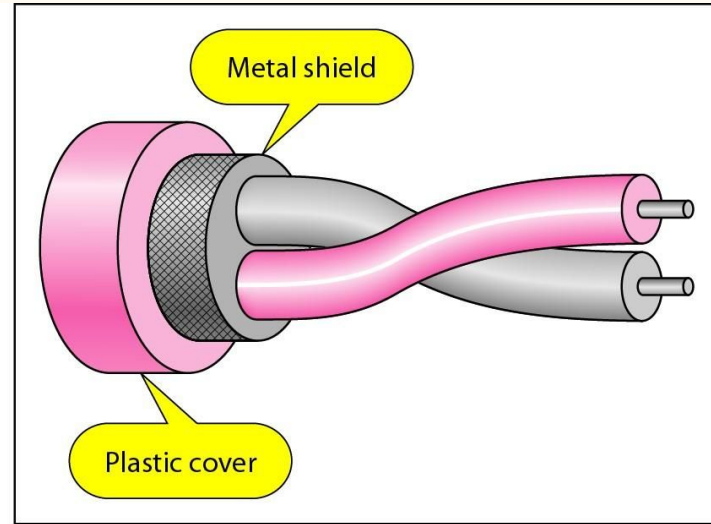
- Susceptibility to noise.
- Least secure guided media.
- Distance limitations – short range.
- Low data rate.
- Requires more expensive hubs.

Guided Media: Types of Twisted Pair

1. Unshielded Twisted Pair
2. Shielded Twisted Pair



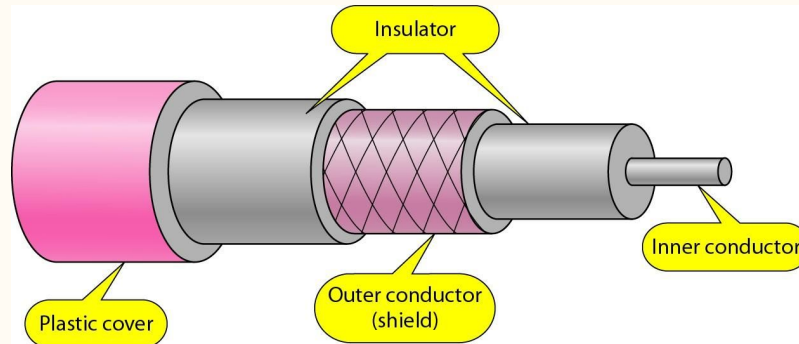
a. UTP



b. STP

Guided Media: Coaxial Cable

- Coaxial cable carries signals of higher frequency ranges than those in twisted pair cable.
- Coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two.
- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating
- sheath, and the whole cable is protected by a plastic cover.



Guided Media: Coaxial Cable

Advantages:

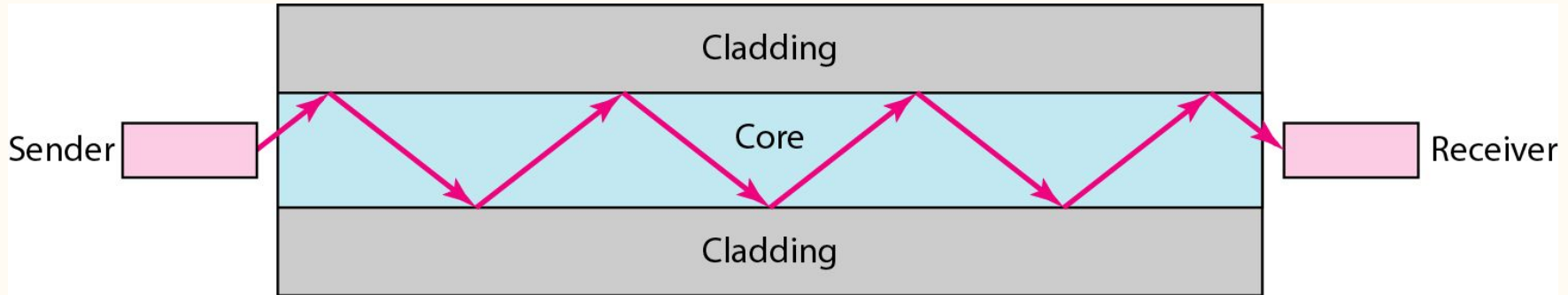
- Bandwidth is high
- Used in long distance telephone lines.
- Transmits digital signals at a very high rate of 10Mbps.
- Much higher noise immunity
- Data transmission without distortion.
- They can span to longer distance at higher speeds as they have better shielding when compared to twisted pair cable

Disadvantages:

- Single cable failure can fail the entire network.
- Difficult to install and expensive when compared with twisted pair.
- If the shield is imperfect, it can lead to grounded loop.

Guided Media: Optical Fiber

- An Optical Fiber is a guided transmission medium made of ultra-pure glass (silica) or plastic that transmits data in the form of light pulses.
- It operates on the principle of Total Internal Reflection (TIR), allowing signals to travel vast distances with extremely low power loss and immunity to electromagnetic interference.

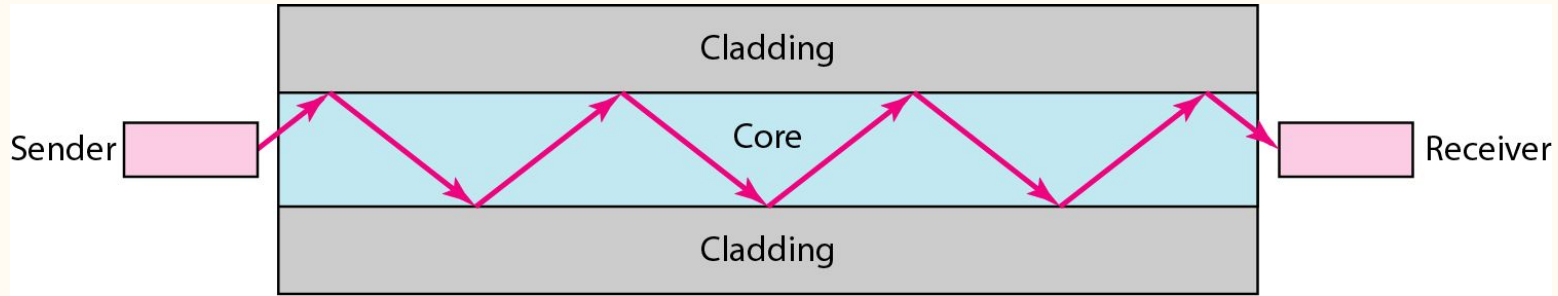


Guided Media: Optical Fiber

Core: The innermost thin strand made of high-quality glass or plastic. This is the actual path where light travels.

Cladding: The outer optical material surrounding the core. It has a lower refractive index than the core, which reflects light back into the core, keeping the signal trapped.

Buffer Coating/Jacket: A protective plastic layer that shields the fiber from moisture, physical damage, and bending stresses.



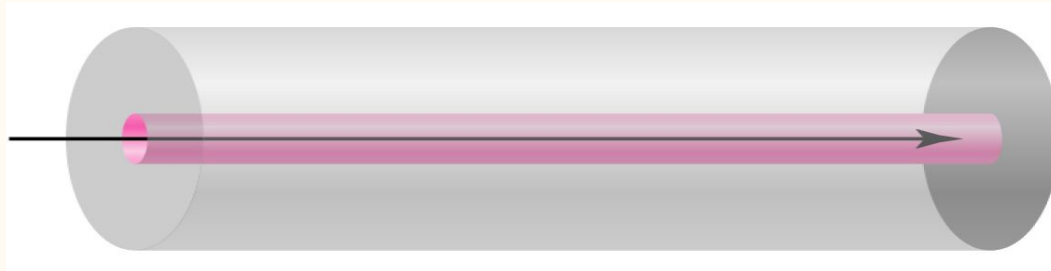
Guided Media: Optical Fiber – Modes

Single-Mode Fiber (SMF):

Mechanism: Allows only one path (mode) of light to propagate.

Application: Used for long-distance communication (e.g., undersea cables, transcontinental lines).

Advantage: Zero "modal dispersion," allowing for the highest speeds and longest distances.



Guided Media: Optical Fiber – Modes

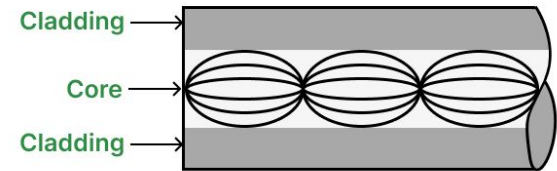
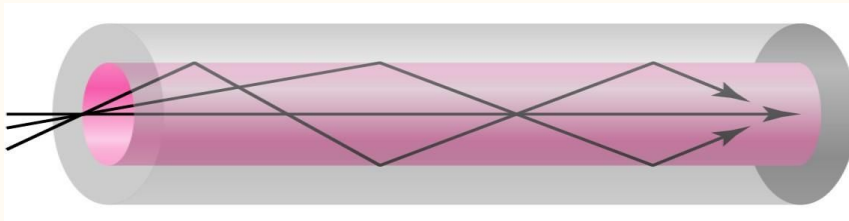
Multi-Mode Fiber (MMF):

Mechanism: Allows multiple paths of light to travel simultaneously. Light bounces off the cladding at various angles.

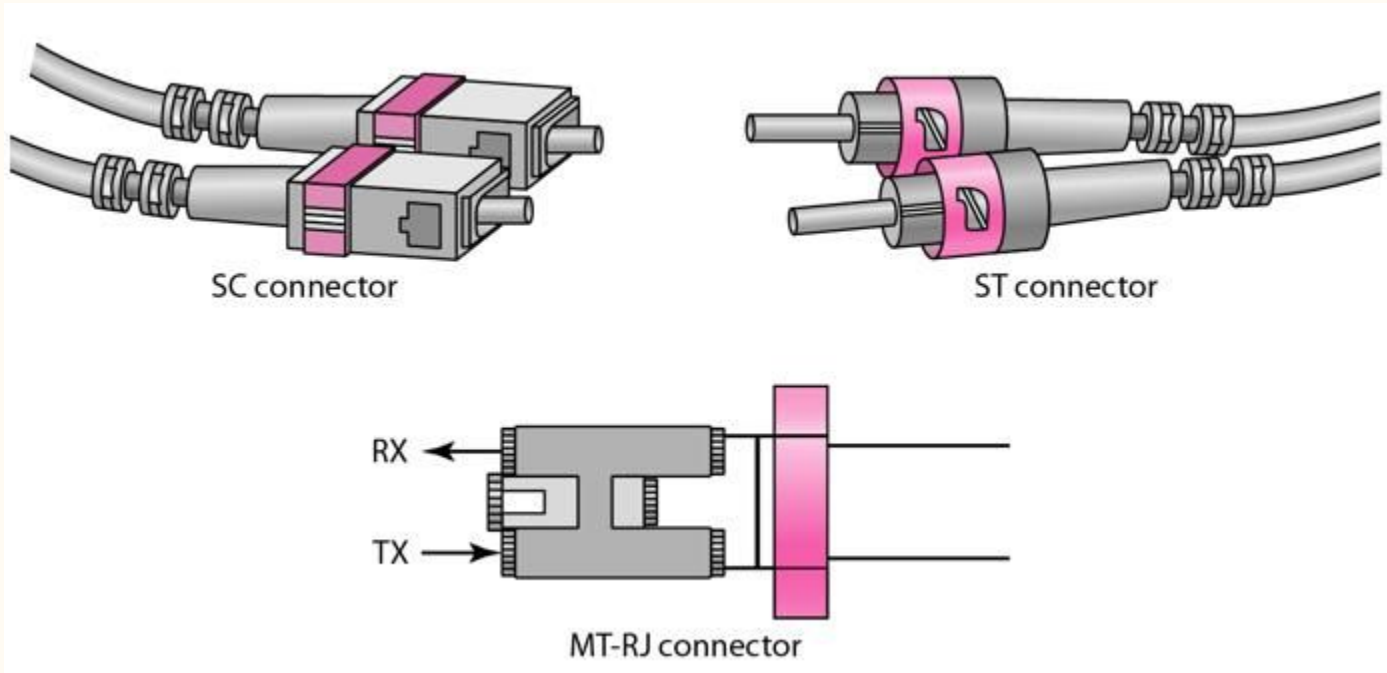
Application: Used for short distances (e.g., within a building or a Data Center).

Disadvantage: Signals "spread out" over distance (modal dispersion), limiting the range.

Type: Step Index and Graded Index.



Guided Media: Optical Fiber Connector



Guided Media: Optical Fiber Connector

Connector Name	Shape	Locking Mechanism	Typical Fiber Count	Primary Application
ST (Straight Tip)	Round	Bayonet (Twist-lock)	Simplex (1 Fiber)	Older LANs, Industrial systems, Campus backbones
SC (Subscriber Connector)	Square	Push-Pull (Snap-in)	Simplex or Duplex	Modern Enterprise Networking, CATV, FTTH
MT-RJ (Mechanical Transfer-Registered Jack)	Rectangular	Latch-tab (RJ-45 style)	Duplex (2 Fibers)	High-density networking, Fiber-to-the-desktop

Guided Media: Optical Fiber

Advantages:

- *Enormous Bandwidth*: Can carry much more data per second than copper cables.
- *Immunity to EMI*: Since it uses light, it is not affected by electrical noise or lightning.
- *Low Attenuation*: The signal can travel for kilometers without needing a booster (repeater).
- *Security*: It is nearly impossible to "tap" into a light signal without breaking the fiber, making it highly secure.
- *Lightweight*: Much thinner and lighter than bulky copper or coaxial cables.

Guided Media: Optical Fiber

Disadvantages:

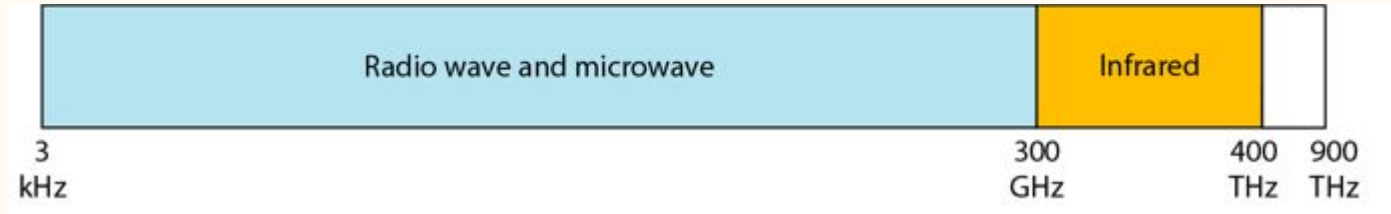
- *High Cost*: The materials and specialized installation equipment are expensive.
- *Fragility*: Glass fibers can fracture if bent too sharply or handled roughly.
- *Complex Splicing*: Joining two broken fibers requires precision alignment and expensive laser-fusion tools.

Unguided Media

Unguided media is a type of transmission medium in which data is transmitted through the air using electromagnetic waves without the use of physical conductors.

Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

Example: Wi-Fi, Bluetooth, radio broadcasting, mobile networks, and satellite communication.



Types of Unguided Media

1. Radio Waves (3 kHz to 1 GHz)

Radio waves spread in all directions from the transmitting antenna.

Characteristics:

- Can travel long distances
- Can penetrate buildings and obstacles
- Do not require precise alignment between transmitter and receiver

Applications:

- Radio broadcasting
- Television broadcasting
- Wireless communication systems
- Maritime and aviation communication

Types of Unguided Media

2. Microwaves (1 GHz to 300 GHz)

Microwaves travel in straight lines, which means the transmitter and receiver must be aligned. This is known as line-of-sight communication.

Characteristics

- High bandwidth and high data transmission capacity
- Requires directional antennas
- Cannot easily pass through obstacles such as mountains or buildings

Applications

- Mobile phone networks
- Wi-Fi communication
- Radar systems
- Satellite communication

Types of Unguided Media

3. Infrared Waves (300 GHz to 400 THz)

Infrared signals are generally used for short-distance communication.

Characteristics

- Cannot pass through walls
- Limited range
- Less interference from other devices

Applications

- TV remote controls
- Wireless keyboards and mice
- Short-range device communication

Unguided Media: Propagation Modes

1. Ground Wave Propagation

In ground wave propagation, radio waves travel along the surface of the Earth.

Characteristics:

- Follows the curvature of the Earth
- Used mainly for low-frequency signals
- Effective for short and medium distances

Applications:

- AM radio broadcasting
- Maritime communication
- Navigation systems

Unguided Media: Propagation Modes

2. Sky Wave Propagation

- In sky wave propagation, radio waves travel upward into the atmosphere and are reflected back to Earth by the ionosphere.
- The ionosphere is a layer of the Earth's atmosphere that contains charged particles capable of reflecting certain radio frequencies.

Characteristics:

- Allows signals to travel very long distances
- Commonly used for international broadcasting
- Works mainly with high frequency (HF) radio waves

Applications:

- Shortwave radio broadcasting
- Long-distance radio communication
- Amateur radio communication

Unguided Media: Propagation Modes

3. Line-of-Sight (LOS) Propagation

In line-of-sight propagation, signals travel directly from the transmitter to the receiver in a straight line.

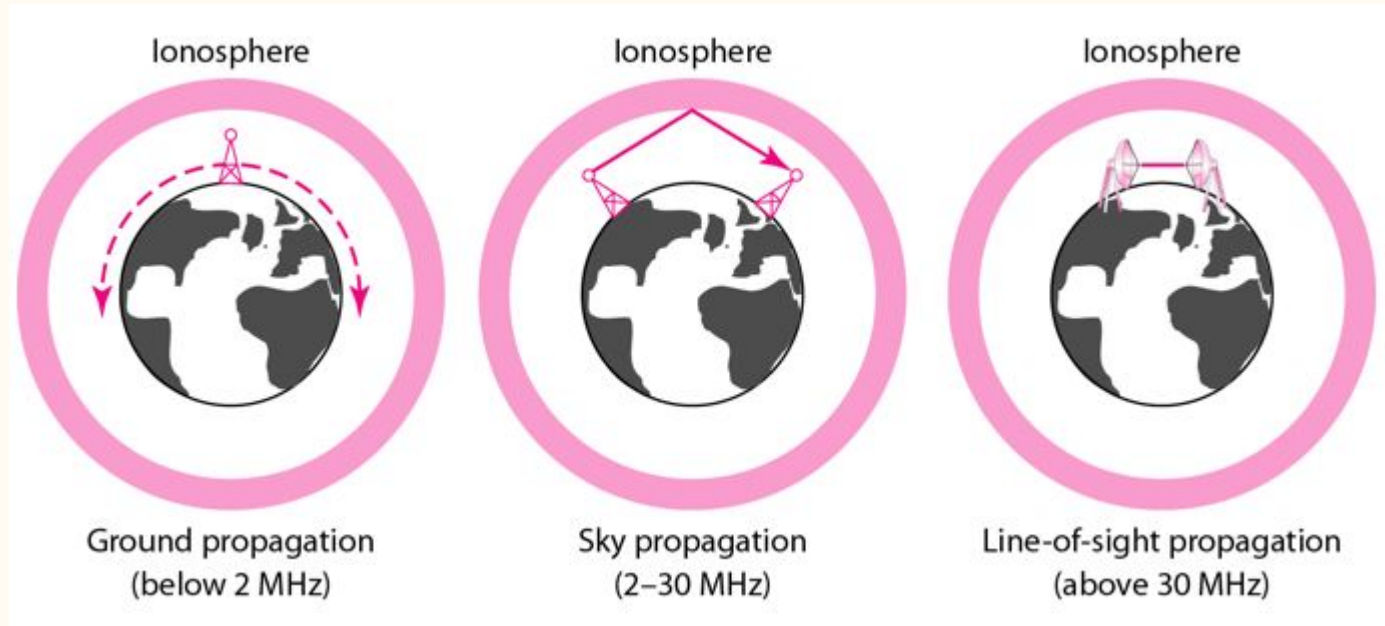
Characteristics:

- Requires a clear, unobstructed path
- Used for high-frequency signals
- Distance limited by Earth's curvature

Applications:

- Microwave communication
- Mobile communication systems
- Satellite communication
- Television broadcasting

Unguided Media: Propagation Modes



Unguided Media: Satellite

Satellite communication is a wireless communication system in which signals are transmitted from an Earth station to a satellite in space, and the satellite retransmits the signals back to another Earth station.

Basic Components of Satellite Communication

1. *Earth Station (Ground Station)*: Ground-based antenna that transmits and receives signals.
2. *Satellite*: A communication device placed in orbit that receives, amplifies, and retransmits signals.
3. *Communication Link*:
 - Uplink - Transmission of signals from Earth to the satellite
 - Downlink - Transmission of signals from the satellite to Earth

Types of Satellite

1. Geostationary Earth Orbit (GEO)

Altitude: about 35,786 km above Earth

Characteristics:

- Moves at the same rotational speed as the Earth
- Appears stationary from the ground
- Covers very large areas

Applications:

- Television broadcasting
- Weather monitoring
- Satellite internet services

Types of Satellite

2. Medium Earth Orbit (MEO)

Altitude: about 2,000 km to 35,786 km

Characteristics:

- Orbits at medium distance from Earth
- Moves relative to Earth's surface
- Several satellites are required for global coverage

Applications:

- Navigation systems
- Positioning services
- Global navigation networks like Global Positioning System.

Types of Satellite

3. Low Earth Orbit (LEO)

Altitude: about 160 km to 2,000 km

Characteristics:

- Orbits close to the Earth
- Moves very quickly around the planet
- Has low communication delay

Applications:

- Earth observation and imaging
- Satellite internet services
- Communication networks such as Starlink.

Bandwidth Utilization

Bandwidth utilization refers to techniques used to make efficient use of the available bandwidth of a communication channel so that multiple signals or users can transmit data simultaneously.

Two major techniques are used:

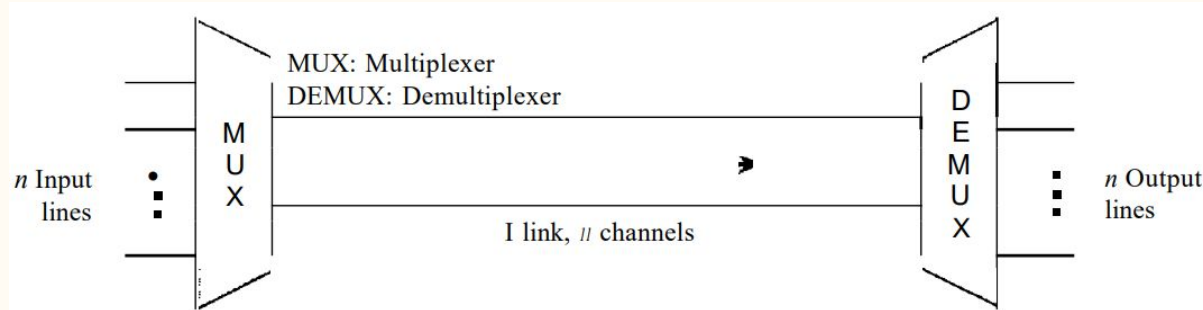
- 1. Multiplexing**
- 2. Spreading (Spread Spectrum)**

Bandwidth Utilization: Multiplexing

Multiplexing is a technique in which multiple signals are combined and transmitted over a single communication channel to make efficient use of bandwidth.

At the receiving end, the signals are separated again using a device called a **demultiplexer**.

Many input signals → Multiplexer (MUX) → Single channel → Demultiplexer (DEMUX) → Original signals



Bandwidth Utilization: Multiplexing

1. Frequency Division Multiplexing (FDM)

In Frequency Division Multiplexing, the total bandwidth of a channel is divided into smaller frequency bands, and each signal is transmitted on a different frequency.

Each user gets a unique frequency range.

Frequency overlap is a serious issue when it comes to frequency division multiplexing and it must be completely avoided. Two frequency ranges can be separated by using some narrow frequency ranges called guard bands.

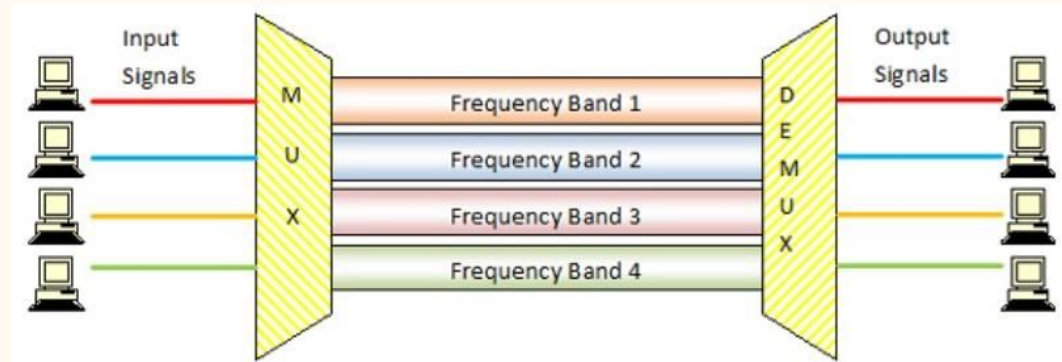
Bandwidth Utilization: Multiplexing

Characteristics:

- All signals are transmitted simultaneously
- Requires guard bands to prevent interference
- Mainly used for analog communication

Applications:

- Radio broadcasting
- Television broadcasting
- Cable TV systems

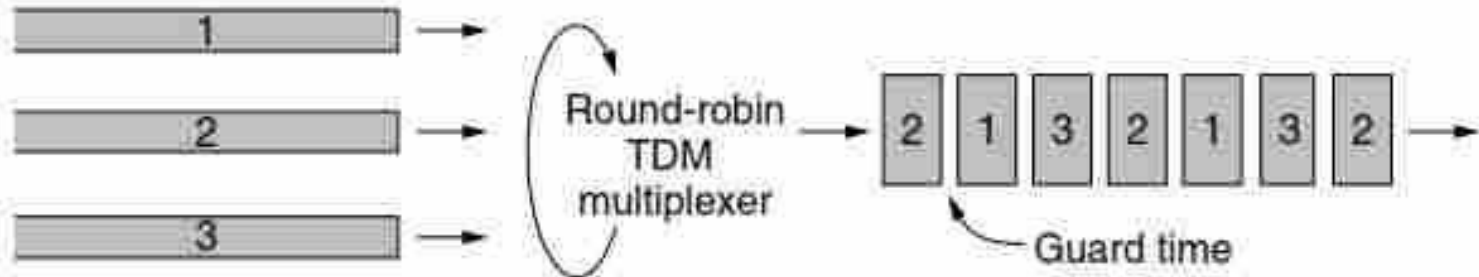


Bandwidth Utilization: Multiplexing

1. Time Division Multiplexing (TDM)

In Time Division Multiplexing, multiple signals share the same channel but transmit in different time slots.

Each user gets access to the full channel bandwidth for a short period of time.



Bandwidth Utilization: Multiplexing

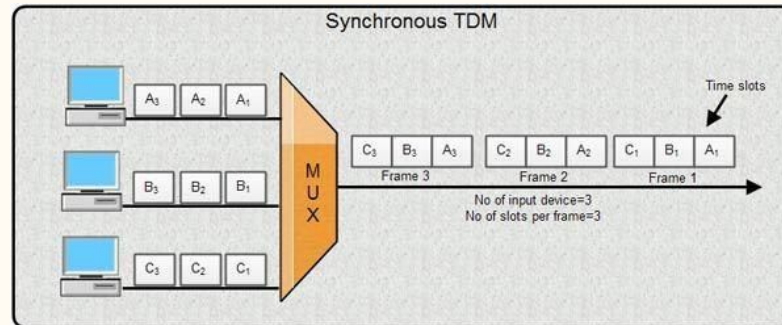
Types of TDM:

a) *Synchronous TDM*

Each device is assigned a fixed time slot even if it has no data to send.

Characteristics

- Simple implementation.
- Time slots may be wasted if a device has no data.



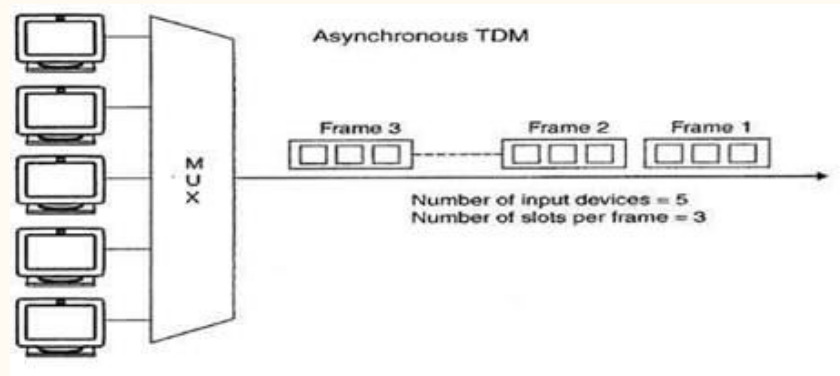
Bandwidth Utilization: Multiplexing

b) Statistical (Asynchronous) TDM

Time slots are assigned dynamically to devices that actually have data to send.

Characteristics

- More efficient bandwidth usage
- More complex control mechanism

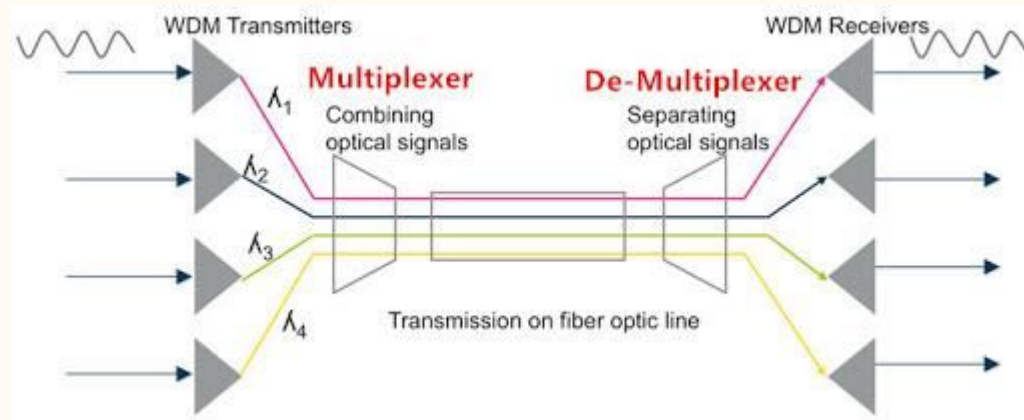


Bandwidth Utilization: Multiplexing

1. Wavelength Division Multiplexing (WDM)

Wavelength Division Multiplexing is used in fiber optic communication where multiple signals are transmitted using different wavelengths of light.

It is conceptually similar to FDM but works with optical signals.



Bandwidth Utilization: Multiplexing

Characteristics

- Extremely high data capacity
- Used in high-speed optical networks

Applications

- Fiber optic communication systems
- High-speed internet backbone networks
- Long-distance telecommunications

Bandwidth Utilization: Multiplexing

Advantages of Multiplexing:

- Efficient use of available bandwidth
- Allows multiple signals to share one channel
- Reduces communication cost
- Increases network capacity
- Supports simultaneous data transmission

Disadvantages of Multiplexing

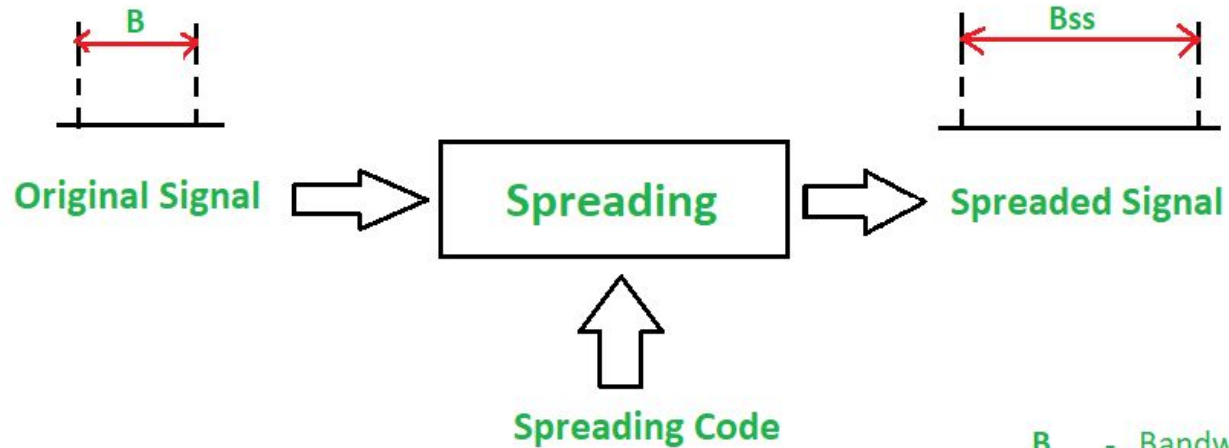
- Requires additional hardware (MUX and DEMUX)
- System complexity increases
- Synchronization may be required
- Signal interference may occur if not properly managed

Bandwidth Utilization: Spreading

Spread spectrum is a communication technique in which a signal is spread over a wide frequency band, much wider than the minimum bandwidth required for transmission.

- Spreading is controlled using a special code sequence.
- Uses a bandwidth much larger than required
- Provides strong resistance to noise and interference
- Improves communication security
- Makes signals harder to intercept

Bandwidth Utilization: Spreading



B - Bandwidth of source signal
 B_{ss} - Bandwidth of spreaded signal

Bandwidth Utilization: Spreading

Frequency Hopping Spread Spectrum (FHSS)

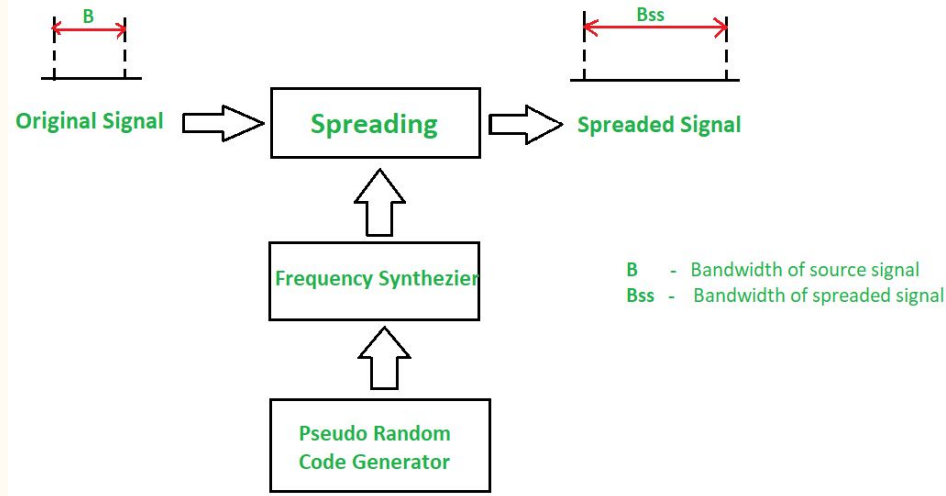
- Frequency Hopping Spread Spectrum (FHSS) is a spread spectrum technique in which the carrier frequency of a signal changes continuously during transmission.
- Instead of transmitting data on a single fixed frequency, the signal hops between many different frequencies.
- A pseudorandom noise (PN) generator creates a pattern of bits.
- This pattern is used to select the carrier frequency from a frequency table.
- The selected frequency is generated by a frequency synthesizer, and the data signal is then used to modulate that carrier frequency.
- After a short time (called the hopping period), the system switches to another frequency based on the next PN code.

Bandwidth Utilization: Spreading

Advantages of FHSS

- Reduces interference and noise
- Provides better security
- Difficult for unauthorized users to intercept the signal

Eg: Bluetooth



Bandwidth Utilization: Spreading

Direct Sequence Spread Spectrum (DSSS)

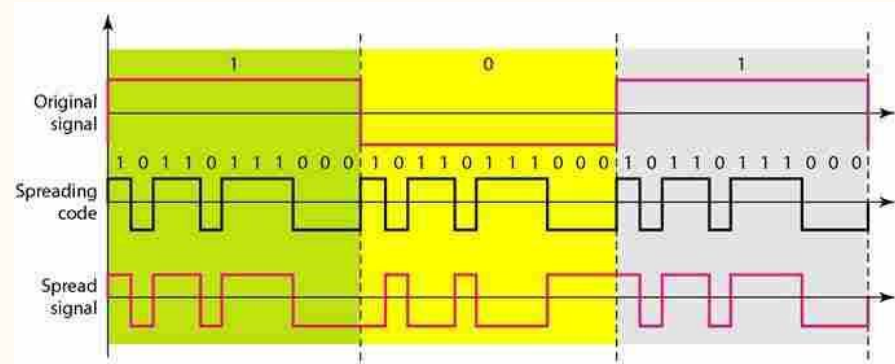
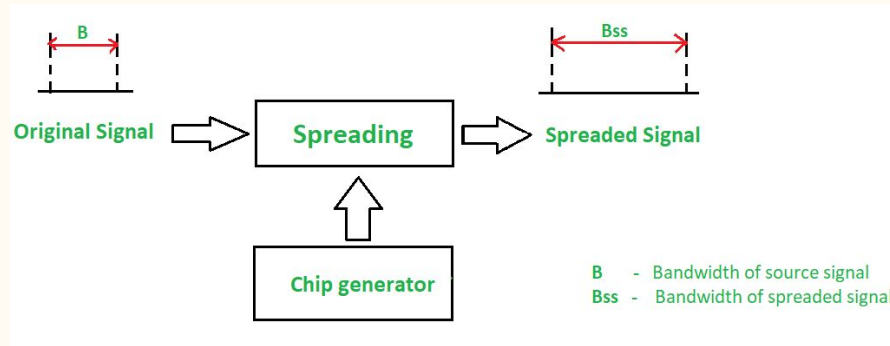
- Direct Sequence Spread Spectrum (DSSS) is a spread spectrum technique in which the original data signal is combined with a high-speed pseudorandom code before transmission
- Each data bit is multiplied by a pseudorandom noise (PN) sequence called a spreading code.
- Because of this, one data bit is converted into many smaller bits called chips.
- At the receiver side, the same PN sequence is used to recover the original data.

Bandwidth Utilization: Spreading

Advantages of DSSS

- Strong resistance to noise and interference
- Provides better security
- Allows multiple users to transmit using different codes

Eg: *Wifi, GPS, CDMA*



Switching

Switching is the process to forward packets coming in from one port to a port leading towards the destination.

When data comes on a port it, is called ingress and when data leaves a port or goes out it is called egress.

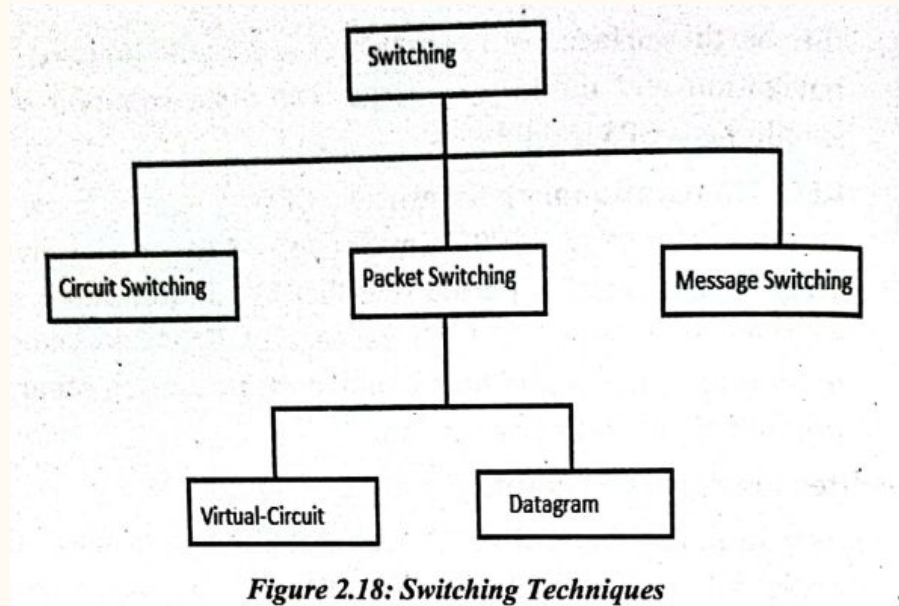


Figure 2.18: Switching Techniques

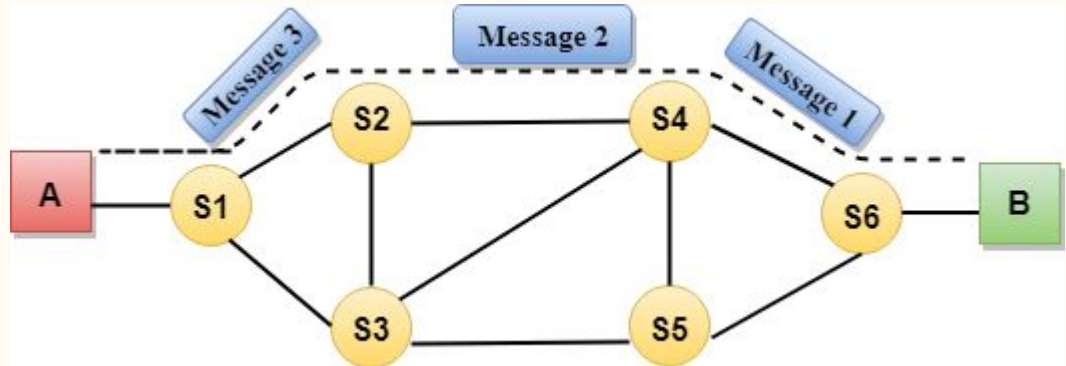
Switching

1. Circuit Switching:

Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.

Steps:

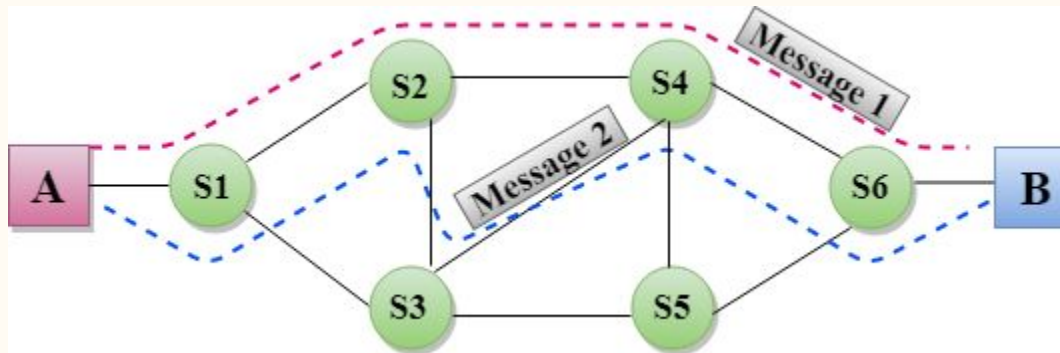
- Circuit establishment
- Data transfer
- Circuit Disconnect



Switching

2. Message Switching:

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- No establishment of a dedicated path between the sender and receiver.
- Also known as **store and forward network**.
- The destination address is appended to the message.



Switching

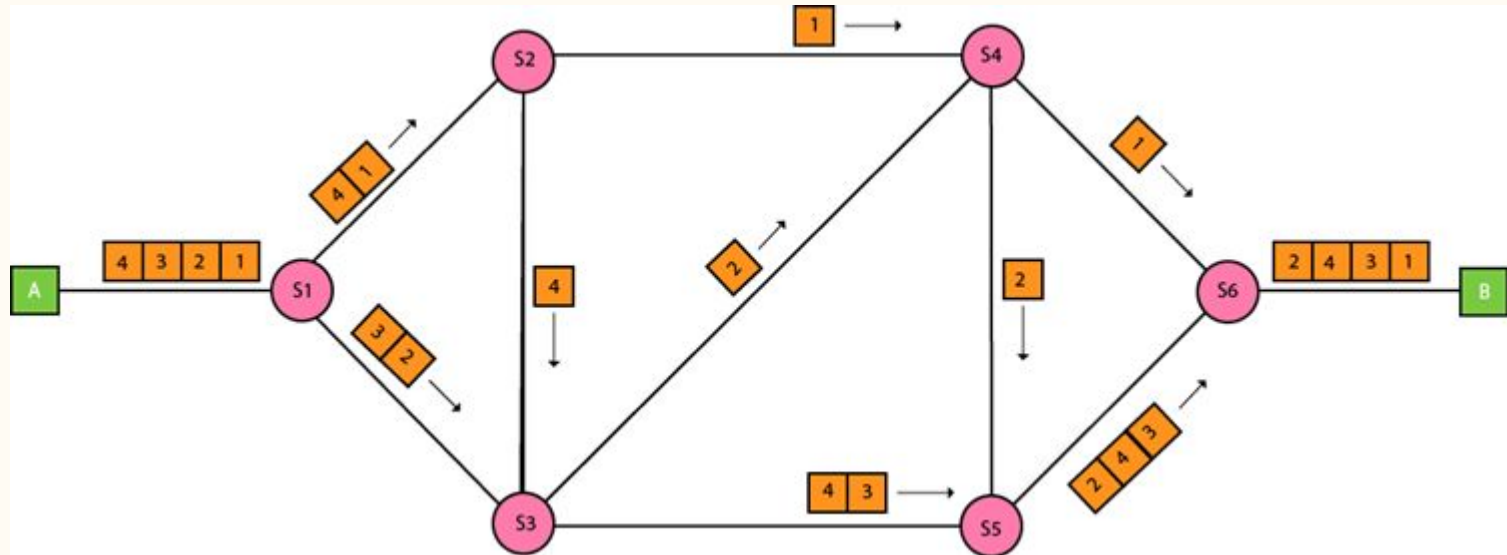
3. Packet Switching:

- The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.

Switching

There are two approaches to Packet Switching:

1. Datagram Packet switching:
2. Virtual Circuit Packet switching



Switching

Datagram Packet switching:

- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity.
- Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- It is also known as connectionless switching.

Virtual Circuit Switching

- Virtual Circuit Switching is also known as connection-oriented switching.
- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.

Telephone Network

- The **Public Switched Telephone Network (PSTN)** is the traditional system used for telephone communication.
- It is a global network made up of copper wires, switching centers, and transmission equipment that allows people to make voice calls over long distances.
- PSTN was designed to carry analog voice signals.
- With the growth of the internet, PSTN has also been used for data communication
- Technologies using the PSTN for internet:
 - Dial Up
 - DSL

Telephone Network: Dial Up

- One of the earliest methods of accessing the internet using the PSTN
- It works by using a device called a modem, which converts digital signals from a computer into analog signals suitable for transmission over telephone lines, and vice versa.

Working:

- The modem dials the ISP's number through the telephone line, similar to making a phone call.
- Once the ISP's modem answers, a connection is established and data transmission begins.
- The receiving modem converts the analog signals back into digital form.
- The connection remains active only while the call is maintained and ends when the user disconnects.

Telephone Network: Dial Up

Advantages

- They use existing telephone lines and do not require additional infrastructure.
- They are widely available, especially in rural or remote areas where other internet technologies may not be present.
- The setup process is simple and does not require advanced technical knowledge or equipment.

Disadvantages

- Very low data transfer speeds, which makes them unsuitable for modern internet usage such as video streaming.
- The telephone line cannot be used for voice calls while the internet is connected, as both services share the same line.
- The connection must be established every time the user wants to access the internet, which wastes time.
- The connection is often unstable and may disconnect due to noise or interruptions in the telephone line.

Telephone Network: DSL

DSL(Digital Subscriber Line) is a technology that provides high-speed internet access using the same telephone lines as PSTN. It transmits data in digital form and allows simultaneous use of voice and internet services.

Working:

- DSL divides the available frequency of the telephone line into separate channels for voice and data.
- A DSL modem connects the user to the service provider without the need to dial a number.
- The connection is always active, unlike dial-up.
- Data is transmitted through a device called DSLAM at the telephone exchange, which connects users to the internet backbone.
- The speed of DSL depends on the distance between the user and the telephone exchange.

Telephone Network: DSL

ADSL

- ADSL (Asymmetric Digital Subscriber Line) is a type of DSL in which the download speed is higher than the upload speed.
- It is designed for home users who typically download more data than they upload.
- It is commonly used for activities such as web browsing, video streaming, and downloading files.

SDSL

- SDSL (Symmetric Digital Subscriber Line) is a type of DSL in which the upload and download speeds are equal.
- It is mainly used by businesses that require consistent data transfer in both directions.
- It is suitable for applications such as video conferencing, online servers, and large file transfers.

Telephone Network: DSL

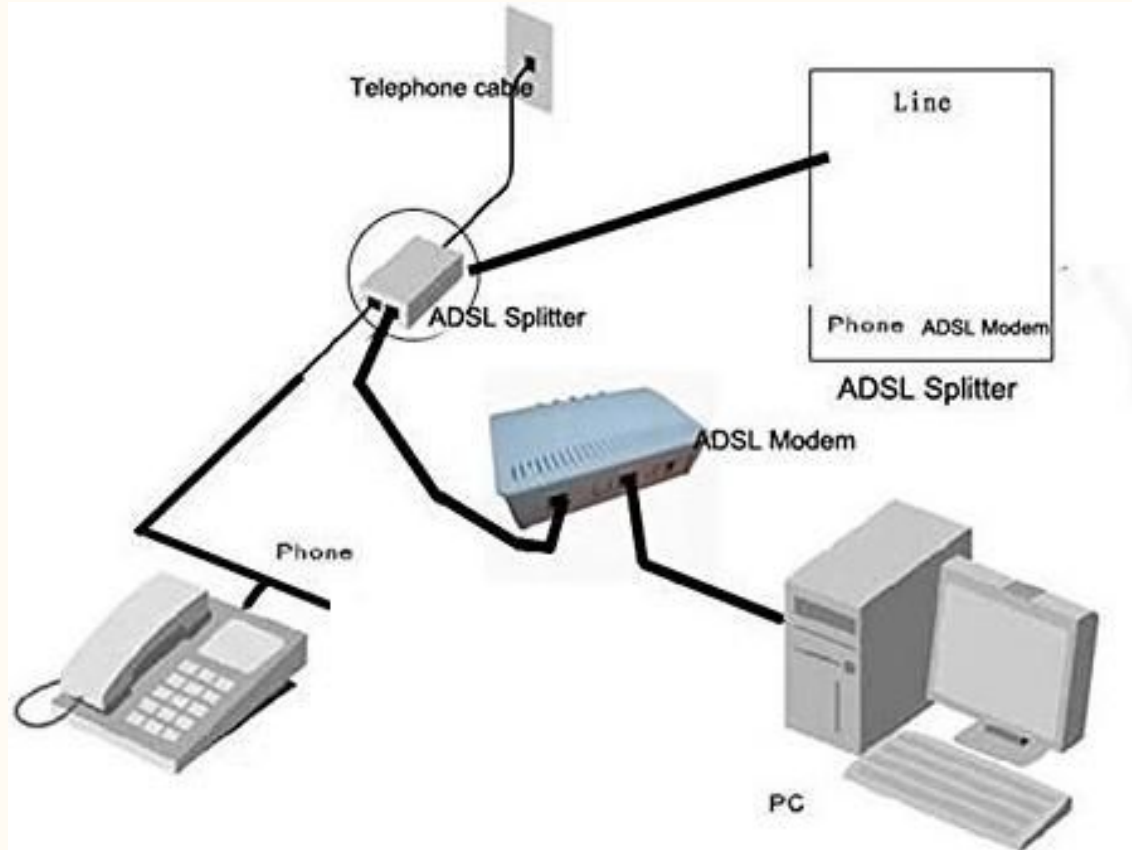
Advantages:

- Faster internet speed than dial-up
- Can use the internet and make phone calls at the same time without interruption.
- Do not need to reconnect every time.
- DSL uses existing telephone lines.

Disadvantages:

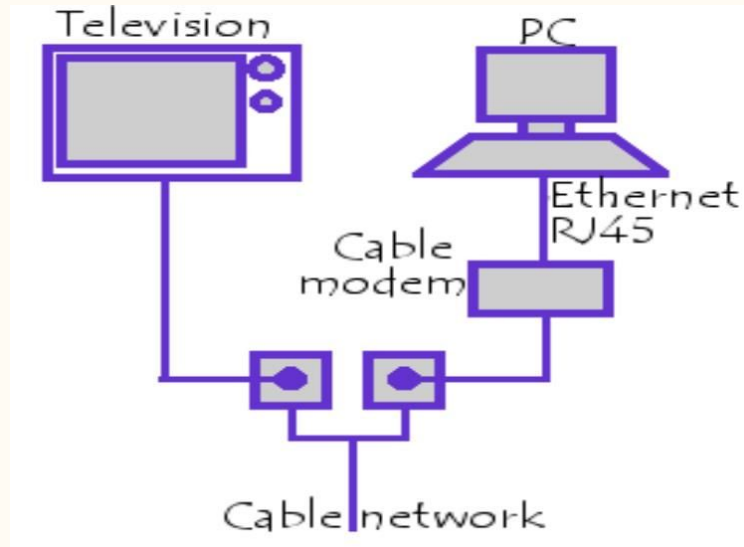
- Speed becomes slower if your location is far from the telephone exchange.
- Slower than modern connections like fiber, especially for high-speed needs.
- Performance depends on the quality of telephone lines, and poor lines reduce speed and stability.
- ADSL has lower upload speed than download speed, which affects uploading files and video calls.

Telephone Network: DSL



Cable Network

- Cable modems provide Internet access using the same cables that transmit cable television
- Cable modems are primarily used to deliver broadband internet access in the form of cable internet



Cable Network

Advantages:

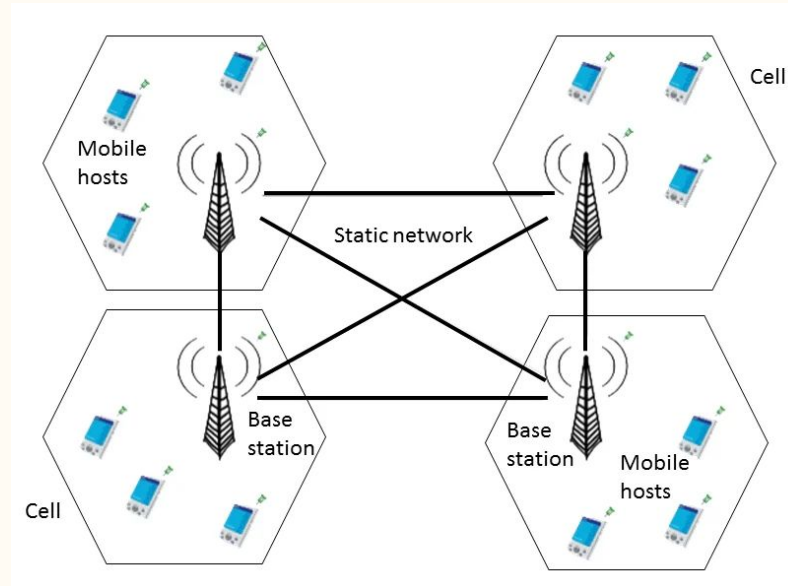
- It provide high-speed internet that supports fast downloading and smooth video streaming.
- It offers a stable connection that remains active without repeated setup or dialing.
- It does not interfere with telephone services because internet and phone signals are carried separately.
- It is easy to install in areas where cable TV infrastructure is already available.

Disadvantages:

- It is more expensive than basic DSL or dial-up services.
- Internet speed can decrease during peak hours because bandwidth is shared among users in the same area.
- Not available in all locations, especially where cable TV infrastructure is not installed.
- Have higher security risks because the connection is shared within a local area network.

Mobile/Cellular Network

A mobile network, also called a cellular network, is a wireless communication system that allows users to make calls, send messages, and access the internet using mobile devices through radio waves and a network of base stations.



Mobile/Cellular Network Working

- The mobile network divides a large geographical area into smaller regions called cells.
- Each cell is served by a base station (cell tower) that communicates with mobile devices using radio signals.
- When a user sends data or makes a call, the mobile phone converts the information into digital signals.
- These signals are transmitted as radio waves from the mobile phone to the nearest base station.
- The base station forwards the data to the Mobile Switching Center (MSC) or core network.
- The core network connects the data to the destination or the internet and sends the response back.
- The base station transmits the received data back to the mobile device.
- When the user moves, the connection is automatically transferred from one cell to another without interruption (handover).

Mobile/Cellular Network Working

Generation	Description	Key Features
1G	First generation of mobile networks using analog signals.	Supports only voice calls with poor quality and no data services.
2G	Second generation using digital signals.	Supports voice calls, SMS, and basic internet (GPRS/EDGE).
3G	Third generation with improved data communication.	Supports mobile internet, video calls, and multimedia services.
4G	Fourth generation based on IP technology.	Provides high-speed internet, HD streaming, and online gaming.
5G	Fifth generation with advanced wireless technology.	Offers very high speed, ultra-low latency, and supports IoT and smart systems.

The End